

ORIGINAL RESEARCH OPEN ACCESS

FedQCNN: A Privacy-Preserving Federated Quantum Convolutional Neural Network for Retinal Image Classification

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Received: 8 April 2024 | **Revised:** 17 June 2025 | **Accepted:** 19 August 2025

Handling Editor: Kumar Gautam

Funding: This study was supported by Ministry of Electronics and Information Technology, Government of India (Meity/AWS/Cohort 2/2023 dated 25th April, 2023).

Keywords: learning (artificial intelligence) | neural nets | quantum computing

ABSTRACT

Quantum machine learning (QML) provides the opportunity for the success of an automated medical diagnostic system due to the effective representation of the solution space by quantum entangled states and faster optimisation through quantum superposition. Preserving medical data privacy is crucial while implementing an intelligent and efficient medical service provider system. The federated machine learning model is not only rich in diversified model experiences but also helps to protect patient data privacy. This paper proposes a secure, intelligent Internet of Healthcare Things (IoHT) application with a federated quantum convolutional neural network (FedQCNN) to classify medically significant retinal image patches. Following the gradient descent algorithm, we executed the model optimisation and performed the global aggregation using the weighted average of the local models' parameters. The proposed system achieves an evaluation accuracy of 96.8% on E-Ophtha retinal image dataset. We suggest over-the-air distributed machine learning with wireless multiple access channels to significantly save the scaled-up radio resource requirements for massive connectivity to ultra dense IoT devices. The research emphasises the significant progress of federated quantum machine learning and its prospects in the coming decades.

1 | Introduction

Deep neural networks (DNN), particularly convolutional neural networks (CNN), are the most prominent machine learning models that performed noticeably in various image classification fields in the last decade. In 2019, Google affirmed 'quantum supremacy' over traditional computing architectures [1]. This century, we expect a significant progress in QML, exploiting the implementation of a parameterised quantum circuit (PQC), even in the currently available noisy intermediate-scale quantum machines. The proposed method highlights a promising

federated quantum machine learning framework for the retinal fundus image classification technique that helps to develop a computer-aided eye diagnostic system. A rapid, steady and privacy-preserved sample classification method will improve the retinal screening process.

The drawbacks of classical deep neural network-based retinal fundus image classification are the requirement of an ever-growing number of samples and high memory and time complexity for effective network learning. On the other hand, a quantum bit is the superposition of two classical bit states (0

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and 1). A system of n qubits is a superposition of $2n$ classical n -bit states, whereas a system of n bits can only represent one n -bit state at a time. Due to the existence of superposition states in a quantum system, a quantum-based approach can handle all possible inputs at the same time. Quantum entanglement can measure the highly sensitive correlation among image features. However, in the noisy intermediate scale quantum (NISQ) era, the opportunity to use high dimensional qubit data and high circuit depth may be limited. But some recent studies [1–3] have shown that limited depth quantum neural networks or parameterised quantum circuits are quite demonstrative compared to a conventional neural network.

Medical data privacy and confidentiality are sensitive issues if we consider the application in the healthcare domain. Federated learning (FL) is a machine learning environment where training from locally stored medical data is executed only in the edge devices and data privacy is assured.

Decentralised and collaborative quantum federated learning (QFL) within the quantum cloud poses an opportunistic framework but requires the support of a wireless network, as the backbone of the whole ecosystem. This intelligent, quantum convolutional neural network-based IoHT solution provides a sustainable way to adopt an efficient, confidential and cost-beneficial model from the patient's perspective.

2 | Related Research

Quantum computers are able to provide exponential speedup to several complex problems, such as finding out prime factors of a large number (Shor's algorithm) or searching the unstructured database (Grover's algorithm). The application of Shor's and Grover's algorithms-based quantum circuits is found in refs. [2, 3], respectively. However, currently available NISQ processors do not have error correction provision [4, 5] and cannot handle a high-dimensional feature space or a high quantum circuit depth [6]. The quantum neural network (QNN) is the quantum counterpart of the classical neural network. With this expansion of the network learning horizon to the quantum domain, we expect a possible performance improvement. In the progress of QML, the contribution of variational quantum circuits (VQC) or parameterised quantum circuits (PQC) is notable. It made executing the quantum neural network algorithm on NISQ devices possible [7]. Some works on the evaluation analysis of PQC as classifier are available in refs. [8–10]. According to Kerenidis et al. [11], quantum learning is a new frontier in image identification, but proper research evaluation is required for large-scale implementation. Beer et al. [12] represented a quantum counterpart of the classical neuron. Here, the authors proposed a quantum feed-forward network with faster optimisation and fewer memory requirements. In ref. [13], Abbas et al. evaluated learning models regarding their learning capabilities. They showed that quantum neural networks perform better on the Iris dataset and achieve more effective dimensions than the classical ones. In ref. [14], Oh et al. showed a performance

improvement of the quantum learning-based method over classical methods for image classification on the MNIST handwritten digit dataset. The intricacies of QNN training have been extensively experimented with in refs. [11–14]. Mathur et al. [15] presented retinal classification with QNN on the MNIST retinal dataset. They reduced the image size to 4 or 8 dimensions. Although with reduced dimensions, medically significant information loss is probable. Two other works on the classification of medical images of quantum neural networks are available in the references by Sengupta et al. [16] and Hussein et al. [17], respectively. Sengupta et al. [16] applied a quantum neural network to chest radiographs and Hussein et al. [17] to CT images to identify the presence of COVID-19. A quantum neural network or PQC application for the retrieval of retinal images is presented in ref. [18].

2.1 | Deep Learning-Based Retinal Segmentation and Classification Works

Deep learning-based methods usually suffer from the extensive training phase based on resource, time and computation with the requirement of a large number of training samples for successful learning. Retinal hard exudate detection [19] using patch-based CNN, diabetic retinopathy classification [20] using CNN, retinal blood vessel segmentation [21, 22] using U-form deep learning architecture, retinopathy classification [23] using CNN and RNN, exudate detection [24] using CNN, pre-trained network, restricted Boltzmann machine etc. are some traditional deep learning-based retinal segmentation and classification works.

2.2 | Retinal Image Classification Methods Using Quantum Machine Learning

We present some recent research works on retinal image classification methods using quantum machine learning (QML) from the viewpoints of the dataset used, reported performance metric values and the limitations in a concise form in Table 1.

2.3 | Federated Learning

Data privacy is a general concern for personal medical records [31]. Medical data, utilised for model training, should not be accessed in an unauthorised way. The federated architecture ensures both data privacy and a diverse learning experience of the machine learning (ML) models. Antunes et al. [32] focus on the federated solutions to healthcare case studies by applying respective ML models. In work by Qin et al. [33], the general overviews of fundamental principles of federated learning, efficient communications for training a learning model in a federated environment and intelligent federated wireless networking applications are present. The application of secured and reliable prediction systems in different domains is ubiquitous in modern societies. The paper [33] discusses the ways of efficient communications for federated machine learning.

TABLE 1 | Survey on retinal image classification methods using quantum machine learning.

Author - year [ref]	Method	Type of model	Dataset	Performance metric	Limitations
De et al. (2024) [25]	Diabetic retinopathy detection using federated quantum neural network	Quantum neural network	E-Ophtha	Accuracy = 89.74%, precision = 96.77%, recall = 81.08%, F1-Score = 89.57%	Expensive quantum h/w, a limited number of qubits, decoherence and noise among qubits, the overhead of quantum encoding
Landman et al. (2022) [26]	Retinal image classification using quantum neural networks	i. Quantum-assisted neural network ii. quantum orthogonal neural network	MedMNIST	Accuracies using quantum-assisted neural network = 80%, quantum orthogonal neural network = 79%	Limitations of quantum h/w
Toledo-Cort et al. (2022) [27]	Diabetic retinopathy grading with deep quantum ordinal regression	Deep quantum ordinal regressor model (DQOR)	EyePACS, Messidor-2	EyePACS dataset: Sensitivity = 87%, specificity = 98%, AUC = 98% and Messidor-2 dataset: sensitivity = 79.74%, specificity = 92.91%, AUC = 92.39%	—
Mir et al. (2022) [28]	Detection of diabetic retinopathy by classical-quantum transfer learning model	Classical-quantum transfer learning	APTOS 2019 blindness detection dataset	Accuracies in: PennyLane = 91.48%, BasicAer = 93.25%, Cirq = 94.11%, classical = 85.14%	—
Naick et al. (2022) [29]	Drusen, choroidal neovascularisation (CNV) and diabetic macular oedema (DME) affected OCT image classification using QML model	Hybrid quantum and classical model	Duke dataset of OCT images	Accuracy = 95%	—
Raja Sankari (2023) [30]	Retinopathy of prematurity detection using quantum machine learning	Quantum support vector machine (QSVM)	Dataset of SRM Medical College and Research Centre, Tamil Nadu, India.	Accuracy = 95.5%	The system cannot detect different stages of ROP

2.4 | Quantum Federated Learning

Yang et al. [34] present a speech recognition application of QCNN and vertical federated learning. But this work utilises QCNN for feature extraction and assumes the presence of a quantum communication network. Narottama and Shin [35] briefly visualise the general QFL architecture for optimising wireless networks. In this work, the access points are responsible for running a shallow QNN model and sharing the learning parameters with a server having another deeper QNN model. However, this work needs to be implemented practically and provides a descriptive perception without any Experimental results. Yamany et al. [36] proposed quantum-behaved particle swarm optimisation for hyperparameter optimisation of

federated learning. Another work of Chen and Yoo [37] combined quantum models with a pre-trained CNN model in a federated environment. Transfer learning is used here with a QNN to make predictions.

2.5 | Over-the-Air Computation (AirComp)

A federated algorithm for global model aggregation tackles challenges such as data privacy preservation and unbalanced data across multiple devices. The weighted average of the locally updated model parameters generates the global model. However, due to the limited communication bandwidth, this method is only sometimes efficient for aggregating the locally computed

updates. Zhu et al. [38] suggest a specific communication method—over-the-air computation (AirComp) that could be leveraged to aggregate model updates in distributed learning applications. Yang et al. [39] utilise over-the-air computation to overcome bandwidth limitations and explore the superposition property of a wireless multiple-access channel for fast global model aggregation. In the proposed work, the characteristic vectors of image patches are obtained as CNN outputs. They are inputs to the QNN circuit to get the output probability at the measurement unit of the circuit. The circuit parameters are optimised using the parameter shift rule. A diversely experienced and efficient federated model is also able to address the concern of medical data privacy. Over-the-air computation of model parameters is proposed to overcome the challenge of the limited bandwidth of the wireless channels. The work exploits the benefits of different research domains. The implementation of the federated model is highlighted and discussed in the following sections. The work does not assume any quantum communication network but utilises a standard classical federated communication framework for parameter transfer.

2.6 | Motivations

This paper proposes a decentralised, federated quantum convolutional neural network for retinal image classification. With some inherent properties of quantum computing (QC) like entanglement, superposition and parallelism, QML will offer efficient pattern identification power with exponential speedup and fewer memory requirements than its classical counterpart. Quantum entanglement captures the sensitive correlation among the image features. Improvements in medical image classification and subsequent analysis would enhance image-aided diagnostics and precise treatment. Although different researchers notably explored retinal classification in classical domain research, the pertinency of applying federated quantum learning in the relevant field, keeping the confidentiality of sensitive medical information unexplored, requires research attention. Collecting a large volume of medically important data is tricky because patient privacy and data security would be at stake, and hospitals cannot avoid their responsibilities. For patient privacy preservation, the distributed learning framework is desirable. The central concept behind over-the-air computation or AirComp is to exploit the wireless channel's superposition property to compute the required function. The AirComp approach effectively estimates the gradient means, resulting in better energy, latency and spectral efficiency when compared to conventional orthogonal communication methods. We suggest employing AirComp in the federated network architecture's design based on this idea. These are the motivations behind the design and implementation of the work.

2.7 | Contributions

The contributions of this work are as follows:

- Implementation and performance analysis of privacy-preserving distributed federated training of the FedQCNN model for retinal image classification.

- Implementing model parameter transfer over classical wireless channels instead of transferring quantum objects or data over quantum channels which is a real-time implementation challenge so far.
- Design of the wireless communication model with over-the-air computation (AirComp) in the distributed machine learning framework. AirComp helps to reduce the server-side computation overhead with reduced radio resources compared to other orthogonal communication methods.

This paper addresses the rising importance of fast and efficient medical data classification and the need for data security and confidentiality.

We organise the article in the following way. Section 3 discusses the preliminary concepts required to present the work. The subsections of Section 4 represent the experiments and results regarding QFL implementation and evaluation, network architecture design etc. Section 5 discusses the work, challenges and potential remedies. Finally, Section 6 concludes the article.

3 | Preliminaries

3.1 | Federated Learning

This variation of distributed learning arises from the recently developed requirements for handling data privacy while a system provides decentralised services. The federated learning system comprises a global node and several edge nodes. In federated learning, edge nodes are responsible for local machine learning model training, and the learnt parameters of edge models are transmitted to the central server node over the network. This central server node aggregates those model parameters to form the global model. These global parameters are sent back to the local client nodes, where further training iterations occur. The evaluation of the final aggregated model takes place with local patient data. In our QNN-based distributed learning system, regional nodes perform the training of quantum models in quantum computers. During training, quantum circuit parameters are updated using the quantum gradient descent (QGD) algorithm applying the parameter shift rule [29]. These updated client parameters are aggregated following the weighted average of parameters at the central node to form a new learning model.

3.2 | Basics of Quantum Computing (QC)

The fundamental element of quantum information is the quantum bit or qubit. One qubit has two basis states, $|0\rangle$ and $|1\rangle$. But the qubit $|\Psi\rangle$ is in the superposition state of both (Equation 1a).

$$|\Psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1a)$$

α and β are complex coefficients. During measurement operation, this superposition state collapses in one of the basis states,

which is unpredictable. At state collapse, the probability of getting one of the two basis states $|0\rangle$ and $|1\rangle$ have probabilities $|\alpha|^2$ and $|\beta|^2$, respectively. Thus, $|\alpha|^2 + |\beta|^2 = 1$. In polar coordinates (θ, ϕ) , the qubit $|\Psi\rangle$ can be represented as (Equation 1b).

$$|\Psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi} \sin\left(\frac{\theta}{2}\right)|1\rangle, \quad 0 \leq \theta \leq \pi, \quad 0 \leq \phi \leq 2\pi \quad (1b)$$

This polar coordinate representation leads to the three-dimensional Bloch sphere representation of the qubit, where the qubit is placed on the surface of the unique three-dimensional Bloch sphere.

An n -qubit system is a superposition of 2^n basis states. These basis states range from $|00\dots00\rangle$ to $|11\dots11\rangle$, both inclusive. An n -qubit register has 2^n basis states and forms a superposition of 2^n state vectors in Hilbert space—a high-dimensional complex vector space—represented as Equation (2).

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle \quad (2)$$

where $x \in \{0,1\}^n$ denotes an n -bit binary string.

Another quantum phenomenon is quantum entanglement, which causes a correlation between qubits. Operation on qubits is characterised by unitary operators (U). The output of the unitary operator is another superposition state. Unitary quantum gate operations are capable of entangling two or more qubits. A detailed discussion is available in ref. [40].

3.3 | Quantum Gates

Following the principles of quantum mechanics, quantum state conversion is unitary. So, the unitary matrix can represent quantum gates. The definition of unitary operator is Equation (3).

$$UU^\dagger = U^\dagger U = I \quad (3)$$

$U^\dagger$ is the adjoint of U and I is the identity operator. The unitary operator is reversible. According to the definition of unitarity, it must have an inverse operator. The inverse operator is the adjoint operator. Unitarity conserves the inner product between two quantum states. Thus, quantum gates are reversible.

This work uses two-qubit coupling gates, R_{xx} and R_{zz} . The XX and ZZ coupling gates (R_{xx} and R_{zz}) are two-qubit parametric $X \otimes X$ and $Z \otimes Z$ interactions. Maximum entanglement is achieved when $\theta = \frac{\pi}{2}$. The general definitions of XX and ZZ entangling gates are given in Equation (4), where $\otimes$ denotes the tensor product.

$$R_{p_j}(\theta_j) = \exp\left(-i \frac{\theta_j}{2} p_j\right), \quad p_j \in \{X \otimes X, Z \otimes Z\} \quad (4)$$

3.4 | Quantum Neural Network (QNN)

When two research approaches using classical computing and quantum computing combine, it opens up a new unrivalled way to explore a more acquiescent computing opportunity, namely hybrid computing. In the same way, a QNN utilises the concepts of both classical neural networks and quantum computing. As a result, we expect this classification approach to be more flexible and robust than the classical neural network alone. In ref. [41], a developmental discussion of quantum logic to quantum deep learning is available.

Quantum neural networks (QNN) are parameterised quantum circuits (PQCs) along with the quantum encoder circuit and the measurement unit as the classical performance evaluator. Following are the steps to design a QNN:

- Qubit encoding
- Parameterised quantum circuit (PQC) design—Ansatz
- Measurement of PQC using computational basis.
- Classifier—Usually, first qubit measurement decides the class labels, +1 or -1, based on the probability of the labels.
- Classical optimisers compute the error or loss and adjust the ansatz parameters to reduce the loss.

Classical data are preprocessed and encoded to qubits to input in the PQCs. Data encoding places the data in the Hilbert space, a high-dimensional complex vector space, complete with respect to the norm defined with the inner product of the vector space. Parameterised quantum circuit (PQC) or variational quantum circuit (VQC) is a quantum circuit that accepts free parameters and applies gate operations over them. PQC has entanglement layers. VQC/PQC is the subclass of variational quantum algorithms (VQA). PQCs are expressive in producing significant inferences in some cases [9, 42–45]. The circuit is composed of quantum rotation gates and entanglement gates. This block is the circuit block. The measurement of a quantum circuit returns a dictionary object. Classical computers observe the measurements at the output end of the circuit. During measurement or observation, a quantum state collapses into one of its basis states. At measurement, the qubit converts to the classical bit. Measurement is usually associated with a single computational basis, the z -basis, as in Equation (5).

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (5)$$

where, $Z|0\rangle = |0\rangle$ and $Z|1\rangle = -|1\rangle$ The Z -basis is also known as the $\{|0\rangle, |1\rangle\}$ basis.

During the measurement of an arbitrary qubit state ($\alpha|0\rangle + \beta|1\rangle$), the probabilities of measuring $|0\rangle$ and $|1\rangle$ are calculated as

$$P_{|0\rangle} = |\alpha\langle 0|0\rangle + \beta\langle 0|1\rangle|^2 = |\alpha|^2 \quad (6)$$

$$P_{|1\rangle} = |\alpha\langle 1|0\rangle + \beta\langle 1|1\rangle|^2 = |\beta|^2 \quad (7)$$

$$|\alpha|^2 + |\beta|^2 = 1 \quad (8)$$

Figure 1 represents the three blocks of a PQC.

QNN optimises parameters through the parameter learning algorithm. Quantum circuits are differentiable; hence, the circuit control parameters are fine-tuned during network training. The parameter updation through multiple training iterations minimises the error. So, VQC or PQC is considered a QNN. Figure 2 is the block diagram representing QNN training. One can perceive the VQC or PQC implementation as an artificial neural network layer design in the context of QML. In this way, PQC is considered a quantum-classical model.

Both healthy and DR-affected image samples are available in the E-Ophtha [46] dataset. Different lesions (exudate and microaneurysm-haemorrhage) are present in unhealthy samples. The corresponding ground truth images are also available

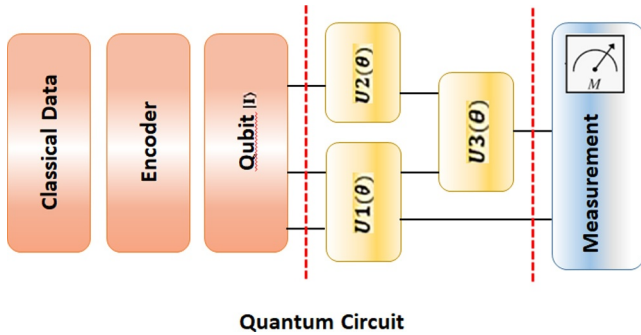


FIGURE 1 | Three blocks of PQC-encoding, circuit and measurement.

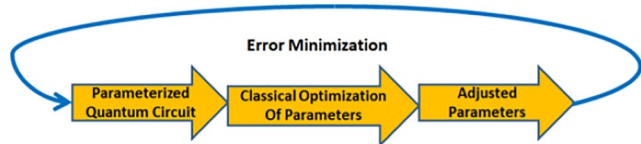


FIGURE 2 | QNN training and classical optimisation of the quantum circuit—A hybrid approach.

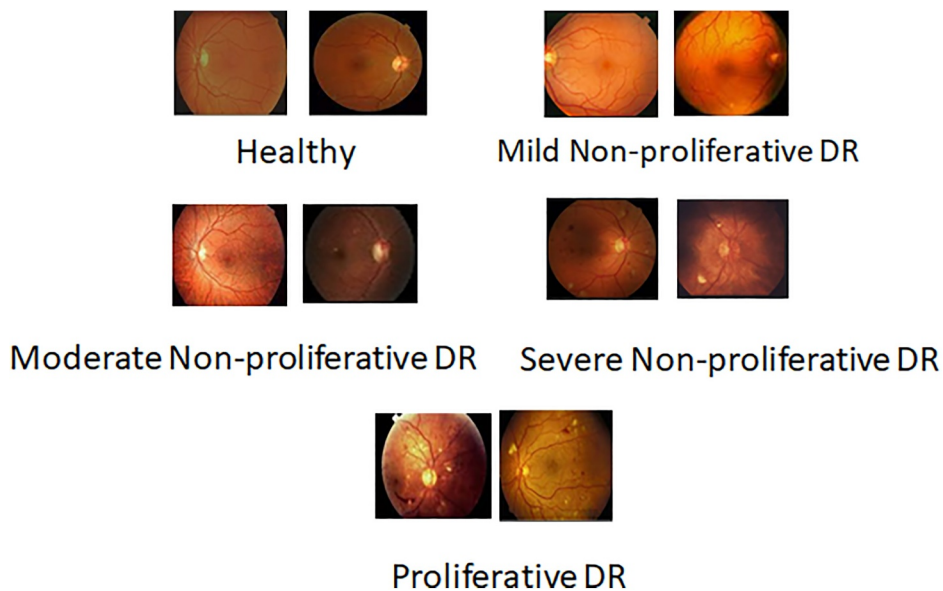


FIGURE 3 | Normal and differently affected sample retinal images based on disease severity stages.

in the dataset. In the proposed application, retinal image samples are captured and pre-processed at the edge nodes such as hospitals, diagnostic centres, health monitors etc. In our implementation, we converted the image samples to green channels and extracted 16×16 patches from these samples. We experimented with an exudate-affected retinal image dataset—E-Ophtha exudate subdatabase. There are 47 exudate-affected images in the dataset. We obtained overlapping 116,365 affected patches and 1,136,505 healthy patches from these images. We experimented with a training database of 5000 affected patches, 5000 healthy patches and a test database of 500 healthy and 500 exudates affected patches. Depending on the number of affected pixels in a patch, we decided whether or not a given patch was healthy. If the number of affected pixels is more than 100, we labelled the sample as unhealthy, and if the number of affected pixels is less than 20, we marked it as healthy.

4 | Experiments and Results

4.1 | QNN Implementation Frameworks

Qiskit is an open-source SDK developed by IBM. We experimented with both IBMQ and AWS real quantum processors, which uses superconducting qubits and different quantum processor simulators used as a backend.

4.2 | Experimental Data

The presence of microaneurysm-haemorrhage (MA-HM) and exudates (EX) in the retina indicate the onset and progress of the retinal disease—diabetic retinopathy (DR) through different ailment stages based on the degree of disease severity, such as mild nonproliferative DR, moderate nonproliferative DR, severe nonproliferative DR and proliferative DR. Figure 3 shows the healthy and differently affected retinal image samples. In DR,

weakened vessel walls are widened from place to place as balloons. These abnormal ballooning are microaneurysms. Eventually, blood soaks out through these damaged and fragile vessel parts. Any fluid seepage through the vessels and deposition of the fluid at the inflammation area is known as exudate.

Both healthy and DR-affected image samples are available in the E-Ophtha [46] dataset. Different image samples are shown in Figure 3. Three blocks of PQC-encoding, circuit and measurement are shown in Figure 2. QNN training and classical optimisation of the quantum circuit are a hybrid approach.

Lesions (exudate and microaneurysm-haemorrhage) are present in unhealthy samples. The corresponding ground truth images are also available in the dataset. In the proposed application, retinal image samples are captured and pre-processed at the edge nodes such as hospitals, diagnostic centres, health monitors etc. In our implementation, we converted the image samples to green channels and extracted 16×16 patches from these samples. We experimented with an exudate-affected retinal

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4.3 | Quantum Encoding and Circuit Implementation

Quantum encoding of classical features is achieved in the following way. In our experiment, the input image patch is 16×16 in size. We extract 4×4 output from successive convolution and max pool layers of CNN as shown in Figure 4. These outputs are flattened to represent 4×4 normalised input vectors. These inputs are angle-encoded with the application of Rx gates. Angle encoding encodes 16 features to 16 qubits.

The circuit implementation is defined with hardware-compatible entangling interactions $X \otimes X$ and $Z \otimes Z$, which are parametric two-qubit rotation operations. It includes 15 XX coupling gates, denoted as R_{XX} , and 15 ZZ coupling gates, denoted as R_{ZZ} . The concise view of the implemented circuit is shown in Figure 5. The XX and ZZ couplings are two-qubit entangling operations and help to create quantum correlations among the qubits. XX coupling represents the interaction along the x-axis, and the ZZ coupling represents the interaction along the z-axis. The combination of XX and ZZ gates combines both amplitude and phase interference. The following is the proof of justification for the implementation of entanglement in the circuit.

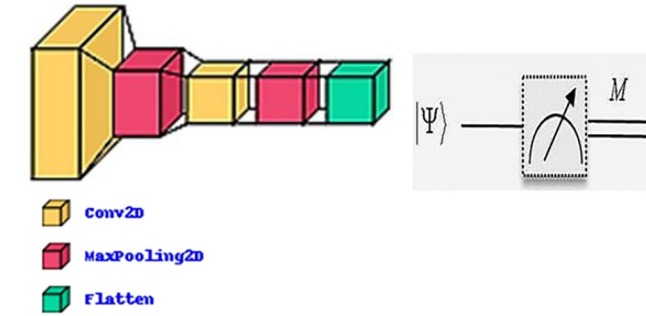


FIGURE 4 | Hybrid FedQCNN model.

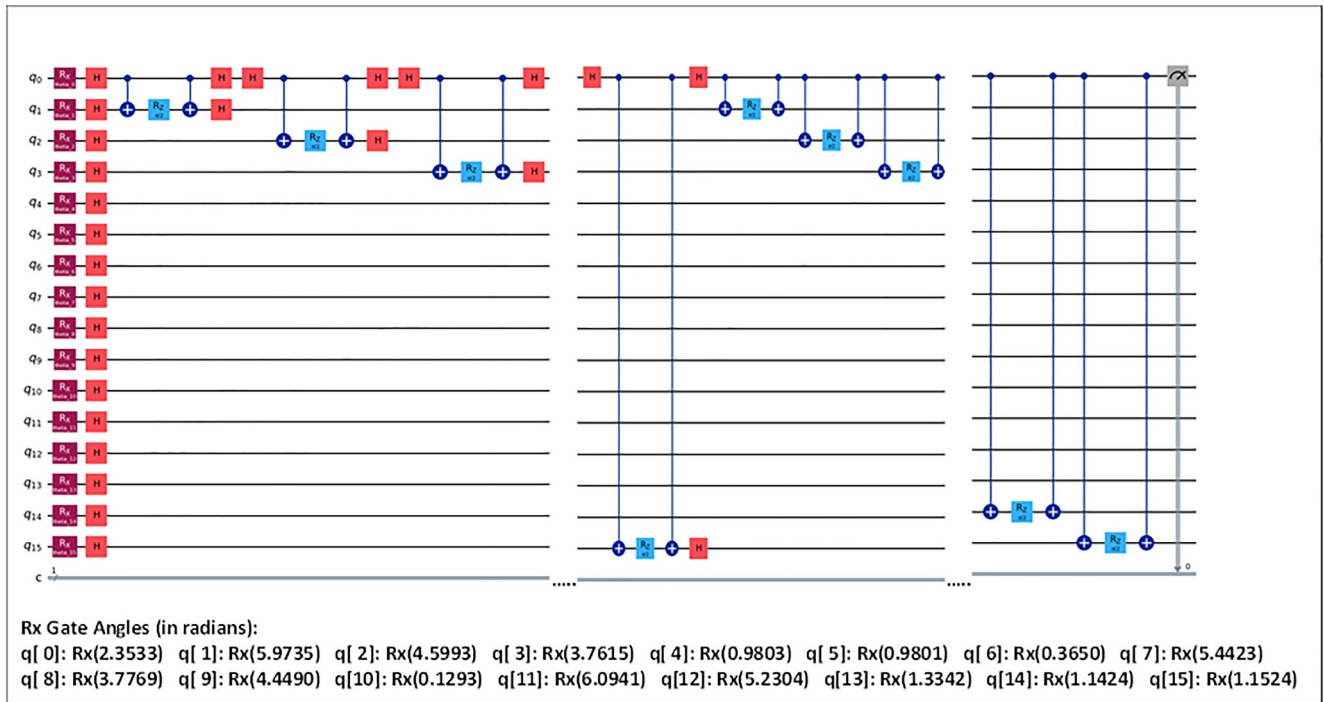


FIGURE 5 | Concise view of parameterised quantum circuit: Part of hybrid QCNN.

To prove

$$e^{-i\theta X \otimes X} = (H \otimes H) \cdot e^{-i\theta Z \otimes Z} \cdot (H \otimes H) \quad (9)$$

We use the identity $HXH = Z$

$$\begin{aligned} &\Rightarrow (H \otimes H)(X \otimes X)(H \otimes H) = Z \otimes Z \\ &\Rightarrow (H \otimes H) \cdot e^{-i\theta X \otimes X} \cdot (H \otimes H) = e^{-i\theta Z \otimes Z} \\ &\Rightarrow e^{-i\theta X \otimes X} = (H \otimes H) \cdot e^{-i\theta Z \otimes Z} \cdot (H \otimes H) \end{aligned}$$

Now, to prove the decomposition of $e^{-i\theta Z \otimes Z}$:

$$e^{-i\theta Z \otimes Z} = \text{CNOT} \cdot (I \otimes R_z(2\theta)) \cdot \text{CNOT} \quad (10)$$

We know

$$\begin{aligned} Z &= \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \quad Z \otimes Z = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ &\Rightarrow e^{-i\theta Z \otimes Z} = \begin{bmatrix} e^{-i\theta} & 0 & 0 & 0 \\ 0 & e^{i\theta} & 0 & 0 \\ 0 & 0 & e^{i\theta} & 0 \\ 0 & 0 & 0 & e^{-i\theta} \end{bmatrix} \end{aligned}$$

Now,

$$\begin{aligned} R_z(2\theta) &= \begin{bmatrix} e^{-i\theta} & 0 \\ 0 & e^{i\theta} \end{bmatrix} \\ &\Rightarrow I \otimes R_z(2\theta) = \begin{bmatrix} e^{-i\theta} & 0 & 0 & 0 \\ 0 & e^{i\theta} & 0 & 0 \\ 0 & 0 & e^{-i\theta} & 0 \\ 0 & 0 & 0 & e^{i\theta} \end{bmatrix} \end{aligned}$$

Then,

$$\begin{aligned} U &= \text{CNOT} \cdot \begin{bmatrix} e^{-i\theta} & 0 & 0 & 0 \\ 0 & e^{i\theta} & 0 & 0 \\ 0 & 0 & e^{-i\theta} & 0 \\ 0 & 0 & 0 & e^{i\theta} \end{bmatrix} \cdot \text{CNOT} \\ &\Rightarrow U = \begin{bmatrix} e^{-i\theta} & 0 & 0 & 0 \\ 0 & e^{i\theta} & 0 & 0 \\ 0 & 0 & e^{i\theta} & 0 \\ 0 & 0 & 0 & e^{-i\theta} \end{bmatrix} \end{aligned}$$

The decomposition, explained above, justifies the XX and ZZ entanglement of Figure 5.

Quantum entanglement is used to capture correlations among image features by encoding individual features into qubits and then applying entanglement gates to these. These entanglement gates generate nonseparable quantum states. Entanglement captures the statistical interdependencies found between different regions of an image, like spatial continuity or discontinuity, repeating or cooccurring textures etc. By entangling specific pairs of qubits, the circuit implicitly learns higher-order feature interactions. The ability of entanglement to jointly

represent features in a high-dimensional Hilbert space offers a rich and compact way to encode complex correlations. In our implementation, entangling one qubit (q_0) with all others makes q_0 a central node of interaction and this connection setup enables q_0 to extract information from the entire circuit.

At the output end, the measurement values are received by the classical processor. For binary classification, only a Pauli operator is measured on the readout, which is the predictor for the QNN. The expectation σ_z is calculated as the circuit measurement, which provides a network output estimate as the average energy of the state.

4.4 | FedQCNN Implementation

Figure 4 shows the block diagram of the FedQCNN model. Figure 6 presents the summary of the CNN part of the proposed FedQCNN model. In the quantum circuit part, we used two-qubit gates. Gates are connected with input qubits and redirected to the readout qubit. The variables in different layers of traditional neural networks are called weights. These weights are circuit parameters in a QNN, that is, parameterised quantum circuits (PQC). Figure 7 describes the proposed hybrid FedQCNN system architecture. FedQCNN combines the power of federated learning (FL) and quantum convolutional neural networks (QCNN).

Image patches are passed through the convolution network layers where the significant image features are extracted as observed in Figure 6. These significant characteristic values are flattened and the input vectors are passed as the parameters to the rotation gates of the proposed quantum circuit. The local model training parameters are transmitted to the global server. The proposed circuit has 16 learnable parameters and 1 bias. The federated aggregation of the local parameters takes place in the global server. The 17 local parameters are aggregated following weighted average aggregation of all the local models as discussed in Section 4.6.

Aggregated QCNN parameters are transmitted back to the local diagnostic centres or hospitals or any other medical service providers etc., where the federated models are fine-tuned with region-specific characteristics. Figure 5 illustrates the implemented PQC. Learning any arbitrary function from some given data is mathematically expressed as optimisation of the network loss function.

4.5 | Optimisation of PQC

Classical optimisation of quantum circuit parameters is a crucial part of PQC training. The quantum computer evaluates the output of the quantum circuit, and a classical optimiser is used to adjust the weights or the parameters. These parameters are the angles passed within the rotation gates and the network bias. After encoding data and executing the quantum circuit on a simulator or real quantum hardware, measurement provides the network outcome, which is compared with the actual outcome to compute the cost or loss function. The classical

Input shape: (None, 16, 16, 1)
Model: "sequential"

Layer (type)	Output Shape	Param #
conv2d_6 (Conv2D)	(None, 14, 14, 4)	40
max_pooling2d_5 (MaxPooling2D)	(None, 7, 7, 4)	0
conv2d_7 (Conv2D)	(None, 5, 5, 4)	148
max_pooling2d_6 (MaxPooling2D)	(None, 2, 2, 4)	0
flatten_3 (Flatten)	(None, 16)	0

Total params: 188
Trainable params: 188
Non-trainable params: 0

FIGURE 6 | Details of convolution part of hybrid FedQCNN.

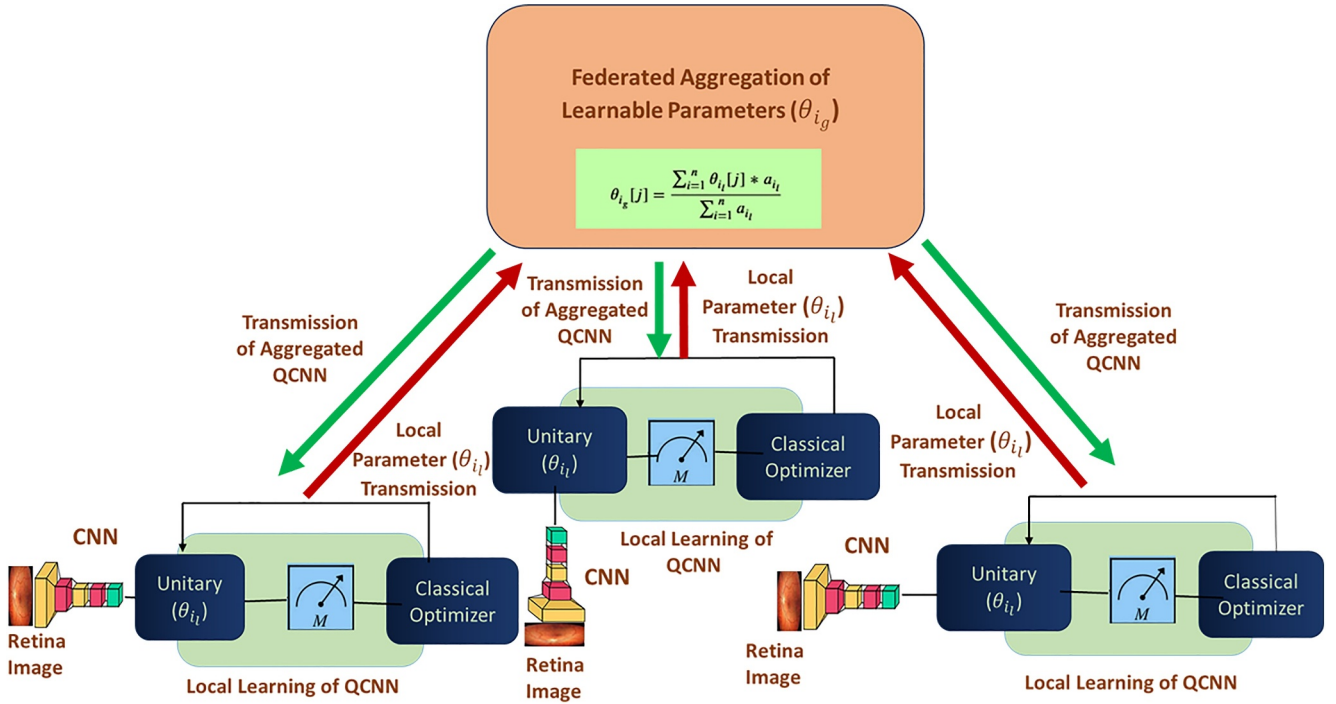


FIGURE 7 | Architecture of hybrid FedQCNN.

optimiser then updates the parameters towards the direction of minimisation of loss. This iterative optimisation loop between quantum evaluations and classical updates enables PQC learning.

In this implementation, network optimisation executes forward propagation and corresponding backward propagation of loss or error. The gradient descent optimisation algorithm computes the gradients of the loss or error function, and the network is updated to minimise the loss or error. Thus, in this case, we require to compute the derivatives of the expectations. The

simple but efficient way to calculate this is to follow the quantum gradient descent algorithm. The algorithm computes the derivatives as a function of the parameter applying the parameter change rule [47]. The derivatives of the expectations are obtained by evaluating the parameterised quantum circuit with parameters shifted in both directions. Thus, Equation (11) computes the derivatives [48].

θ is the parameter to the parameterised gate, and $O(\theta)$ is the expectation of the arbitrary observable O . The derivative of $O(\theta)$ is

$$\frac{\partial \langle O(\theta) \rangle}{\partial \theta} = \frac{1}{2} \left[\langle O\left(\theta + \frac{\pi}{2}\right) \rangle - \langle O\left(\theta - \frac{\pi}{2}\right) \rangle \right] \quad (11)$$

Pauli gate applies rotation to the qubit about an axis in the Hilbert space. Thus, the expectation value is the function of the Pauli gate parameter and can be expressed as a sine function [49], as represented in Equation (12).

$$\langle O(\theta) \rangle = \alpha \sin(\theta + \beta) + \gamma \quad (12)$$

here, α is the amplitude, β is the phase and γ is the displacement. Now, Equation (13) represents the derivative of Equation (12).

$$\frac{\partial \langle O(\theta) \rangle}{\partial \theta} = \alpha \cos(\theta + \beta) \quad (13)$$

Now, a phase shift of $\pi/2$ also converts the equation of sine (Equation 12) to cosine. Thus, we obtain Equation (14),

$$\langle O\left(\theta + \frac{\pi}{2}\right) \rangle = \alpha \cos(\theta + \beta) + \gamma \quad (14)$$

A phase shift of $\pi/2$ to the opposite direction converts the equation of sine (Equation 12) to negative cosine (Equation 15).

$$\langle O\left(\theta - \frac{\pi}{2}\right) \rangle = -\alpha \cos(\theta + \beta) + \gamma \quad (15)$$

We can remove γ by subtracting Equation (15) from Equation (14). The subtraction result (Equation 16) is the same as we obtained in Equation (12), except for a factor of 2 in the front.

$$\langle O\left(\theta + \frac{\pi}{2}\right) \rangle - \langle O\left(\theta - \frac{\pi}{2}\right) \rangle = 2\alpha \cos(\theta + \beta) \quad (16)$$

Thus, factor 1/2 is required in the parameter shift rule of Equation (13) to nullify 2. This is the finite difference method of gradient optimisation. The general parameter shift rule in QML is discussed in ref. [43].

We assigned the network training (forward and backward passes) and evaluation jobs to the actual quantum computer. The quantum circuit execution is allocated to the real quantum computer and deallocated from there to execute the classical part. This phenomenon takes place frequently. Thus, significant time is spent waiting inside the job queue associated with that particular quantum computer both in the IBMQ cloud and AWS Braket cloud.

4.6 | FedQCNN Training and Evaluation

In this study, we introduce a threshold-based labelling criterion to enhance the learning process of the neural network. The samples with more than 100 affected pixels are labelled as unhealthy, whereas those with fewer than 20 affected pixels are considered healthy. This clear demarcation helps the model learn the distinct symptomatic characteristics associated with each class more effectively during the training phase, reducing ambiguity in the representation of features and improving class separation. By focusing on well-defined examples, the network

can better capture the underlying patterns of healthy and unhealthy cases. However, during the evaluation or inference phase, no such strict discrimination is applied. All samples are processed by the trained model to assess its generalisation capability across the entire range of clinical presentations.

During system implementation, the local models are trained for 20 epochs on edge quantum nodes using the regional node's private training data. We put the learning rate (η) to 0.001. The objective of the local training is to minimise the local training loss. Training loss decreases as the training proceeds through epochs. Algorithm 1 represents the local QNN learning procedure.

ALGORITHM 1 | Quantum Federated Model: Local QNN Learning.

```

1: Input: Number of local nodes  $n$ , QNN model object at local
   nodes  $W_i$ , local iteration  $j_i = 1$ , learning rate  $\eta$ , loss  $\epsilon$ 
2: Output:  $n$  number of local QNN models
3: for  $i = 1$  to  $n$  do
4:   Training of local QNN model on local medical data at
   the  $i^{\text{th}}$  location.
5:   while  $\Delta\epsilon$  do
6:     //Weight adjustment using quantum gradient
     descent:
7:      $W_i = W_i - \eta \nabla \epsilon$ 
8:      $W_i = W_i^{i+1}$ 
9:      $j_i = j_i + 1$ 
10:    //Check for local convergence up to no
    improvement of loss.
11:     $\Delta\epsilon = \epsilon_j - \epsilon_{(j-1)}$ 
12:  end while
13: end for
14: for  $i = 1$  to  $n$  do
15:   Local QNN model object  $W_i$  is transmitted to the
   aggregator cloud through a wireless network.
16: end for

```

ALGORITHM 2 | Quantum Federated Model: Global QNN Learning.

```

1: Input: Number of local nodes  $n$ , local QNN model object at
   global nodes  $W_i$ , global iteration  $i_g = 0$ , global QNN model
   object at global nodes  $W_i^g$ , learning rate  $\eta$ , target accuracy  $Acc$ 
2: Output: Trained aggregation model at local nodes
3: Initialise  $W_i^g = \frac{1}{n} \sum_{i=1}^n W_i$ 
4: for  $i = 1$  to  $n$  do
5:   Global QNN model object  $W_i^g$  transmitted back to
   different local medical service providers' locations through the
   wireless network.
6: end for
7: for  $i = 1$  to  $n$  do
8:   Training of returned back global QNN at the  $i^{\text{th}}$  local
   node.
9:   Loss ( $\epsilon$ ) computation.
10:  Global weight adjustment using quantum gradient
   descent:
11:   $W_i^g = W_i^g - \eta \nabla \epsilon$ 
12:   $W_i^g = W_i^{g(i+1)}$ 
13:   $i_g = i_g + 1$ 
14:  Check for convergence up to the desired  $Acc$ .
15: end for

```

Experimental data show that the model training becomes steady with increased training epochs. We experimented with circuit training convergence with epochs and received increasing accuracy curves with variable numbers of epochs. In our approach, the trainings of local models and the federated (global) model follow distinct convergence criteria, as defined by Algorithm 1 and 2, respectively. For local model training, convergence is monitored based on the stabilisation of the loss value and the training continues until there is no further improvement in loss in consecutive training epochs. The condition ensures that each local model reaches its optimal performance on the respective local dataset. The criterion for no improvement of loss is usually defined using a small threshold value denoted as $\Delta\epsilon$, which quantifies the minimum and insignificant change in the loss value. If the absolute difference between the loss values of two successive epochs is less than $\Delta\epsilon$, it is assumed that the model has reached convergence. The value of $\Delta\epsilon$ is chosen from observation and experimental evidence as 10^{-3} . In contrast, the global or federated model training monitors convergence based on achieving a predefined target accuracy, reflecting the overall performance across all participating clients. Due to different stopping criteria, the number of epochs required for local training and global federated training may vary. This adaptive strategy ensures efficient resource utilisation at the local level by allowing each client to stop training once its model has converged or when there is no further improvement in the local loss, instead of enforcing a fixed number of epochs. This prevents unnecessary computation and conserves energy and processing resources at the edge devices, which are generally capacity-constrained. At the federated level, we continue the global training until a desired accuracy is reached instead of being tied to a local training duration. In this way, the model benefits from aggregating diverse and well-optimised local models. This improves the generalisation capability of the global model and enables it to perform robustly on a wide range of data distributions.

Local model training and validation accuracy versus epochs are shown in the Figure 8 and federated model training and validation accuracy versus epochs are shown in the Figure 9. We computed the accuracy of the FedQCNN system considering the number of truly identified affected image patch (True Positive:

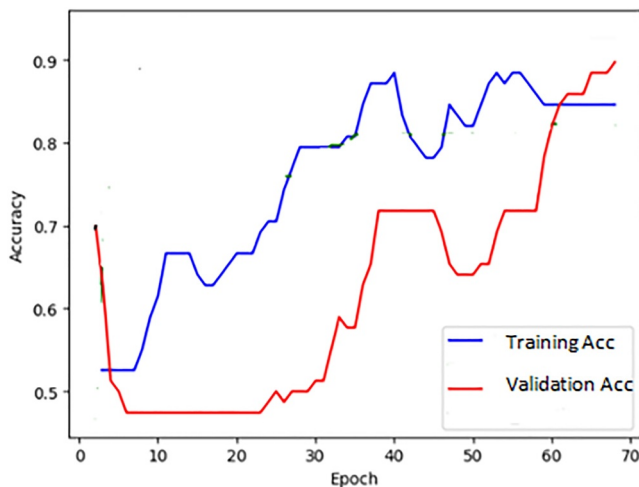


FIGURE 8 | Local training and validation accuracy versus epochs.

TP), truly identified healthy image patch (True Negative:TN), falsely identified healthy image patch (False Positive:FP) and falsely identified affected image patch (False Negative:FN). The computed metric is obtained following Equation (17). To experiment with the trend in quantum training performance with the number of training samples, we generated different prototype training models with 10, 50, and 100 training samples. We evaluated these prototype models with the same test image set. We observed an increase in test evaluation accuracy with increasing train samples as shown in Table 2. The model uses the quantum gradient descent algorithm.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

Efficient and secure aggregation of local model enables initialising the global model. The objective of global training is to minimise global loss. At training start-up, all the 16 parameters of each edge model have a single fixed initialisation value and the weight updation for all the parameters starts from that value. But the parameters or weights of the global model are initialised with the weighted averages of the 16 individual local parameters of all the local models. Hence, global model weights are initialised with different values and weight updation starts from different values for different parameters. Weight initialisations of local and global models are represented in Figure 10. Weight optimisations of local and global models are represented in Figure 11. Global model parameter initialisation and updations are plotted in a graphical form through 10 learning epochs in Figure 11. The global weight updating starts from different

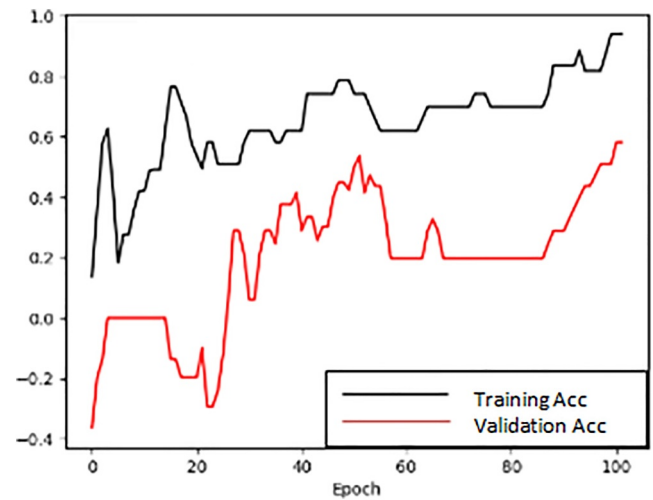


FIGURE 9 | Federated training and validation accuracy versus epochs.

TABLE 2 | Local QCNN performances with the increasing number of training samples.

Number of training patches	Test evaluation accuracy (1000 test samples)
10	40.0%
50	62.0%
100	73.0%

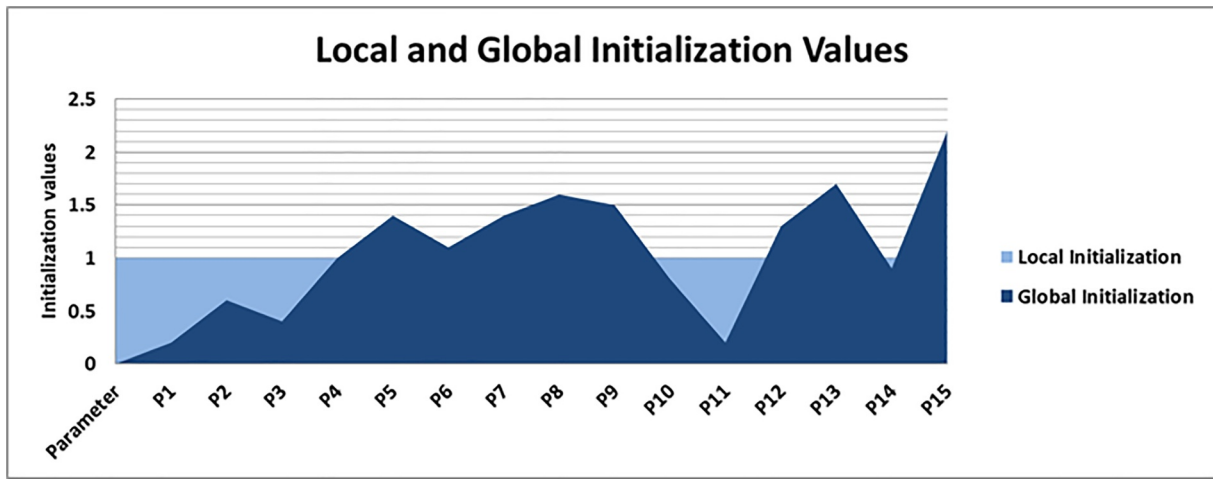


FIGURE 10 | Local and global parameters initialisations in FedQCNN.

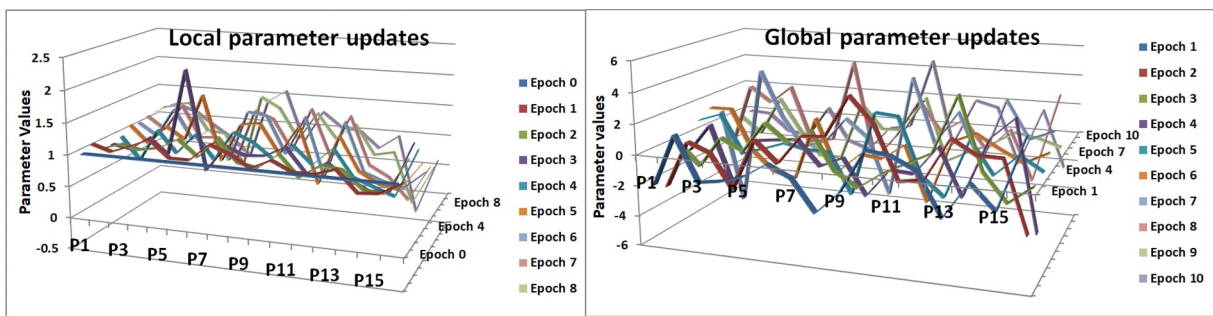


FIGURE 11 | Local and global parameters optimisations in FedQCNN.

weight values, as shown in this figure. Algorithm 2 represents the global QNN learning procedure.

FedQCNN maintains data privacy by implementing federated learning, where raw data are not shared over the network, instead only the learnable parameter values are shared to the federated server. These parameter values are aggregated there and returned back to the local clients. In the local clients, the aggregated models are further trained with region-specific data. This iterative process enhances both robustness and efficiency while ensuring that sensitive medical data remain within the local institution. Thus, the privacy and security of medical data are preserved.

In addition, we experimented with other optimisation algorithms, such as stochastic gradient descent (SGD) and adaptive gradient descent (Adagrad). We plotted the training epoch versus loss curves for these three optimisation algorithms in Figure 12, showing the optimal performance of the Adam optimiser. Figure 13 presents a comparative view of local edge and global server loss-accuracy plots. We executed the experiments on a 16-qubit real quantum machine, though recently, the free public service access of this machine has been withdrawn by IBM. However, the 16-qubit machine was free in the IBMQ platform during the experiment. We also executed the job in IonQ Aria1 in AWS Braket. The credits for research work are available under the MeitY QCAL project. The distributed model learning framework has the capability of preventing medical

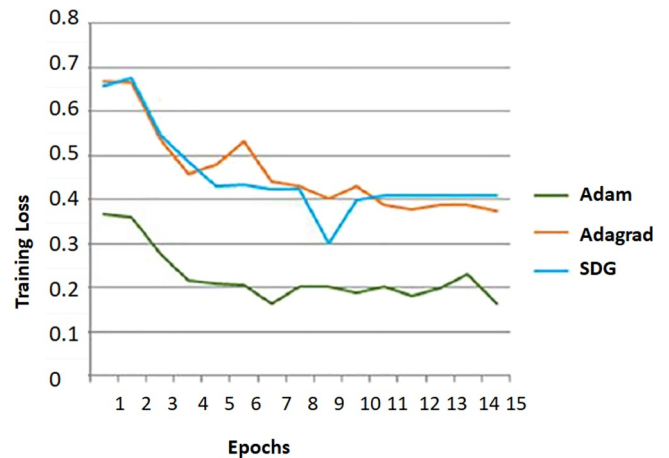


FIGURE 12 | Comparative training convergence with three different optimisation algorithms—Adam, Adagrad and SGD.

information breaches. During weighted average calculation, a better-trained model deserves more weightage than comparatively poorly trained models and contributes accordingly to the aggregated value. Each local model's contribution depends upon the model evaluation performance and varies in our global model aggregation.

Equation (18) represents the weighted average of the i th model parameter or model weight of n local models as

$$W_i = \frac{\sum_{j=1}^n w_j \cdot x_{ij}}{\sum_{j=1}^n w_j} \quad (18)$$

W_i is the i th parameter weighted average, x_{ij} is the i th parameter value of the j th local model and w_j is the weight associated with the j th local model.

It is observed from the test accuracy results that federated learning makes the global model knowledge enriched with diverse local model experiences. Comparative local and global loss-accuracy values are shown in Figure 14.

4.7 | Communication in FedQCNN: Over-the-Air Decentralised Computation

Federated learning involves a wireless aggregation of updated model parameters between the central server and the local clients during the training process. This communication process can consume significant amounts of resources, especially when a large number of local clients are wireless-connected IoT devices, as in Internet-of-Health-Things (IoHT) applications. When multiple local clients transmit their updated parameter sets simultaneously through the same wireless channel, their signals are combined or superimposed in the air, resulting in an

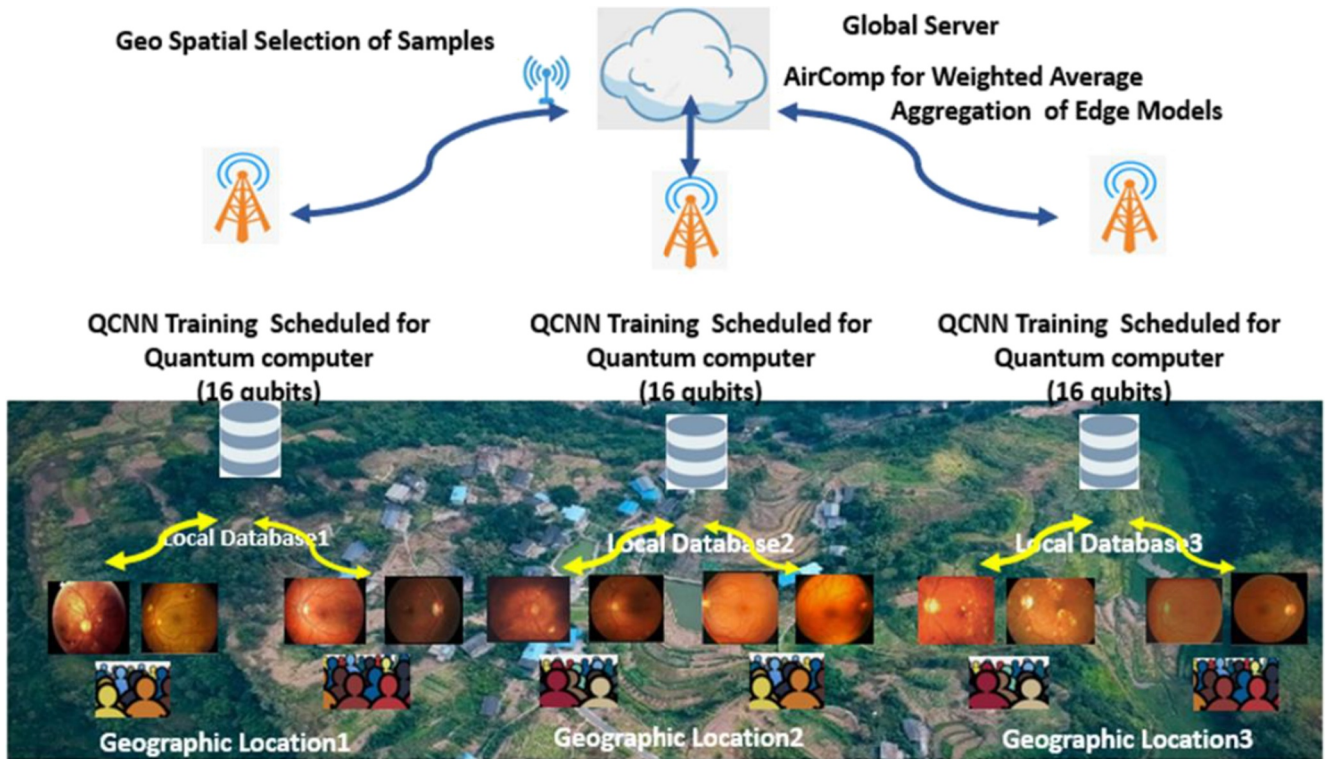


FIGURE 13 | The layered ecosystem of privacy-preserving decentralised quantum federated learning.

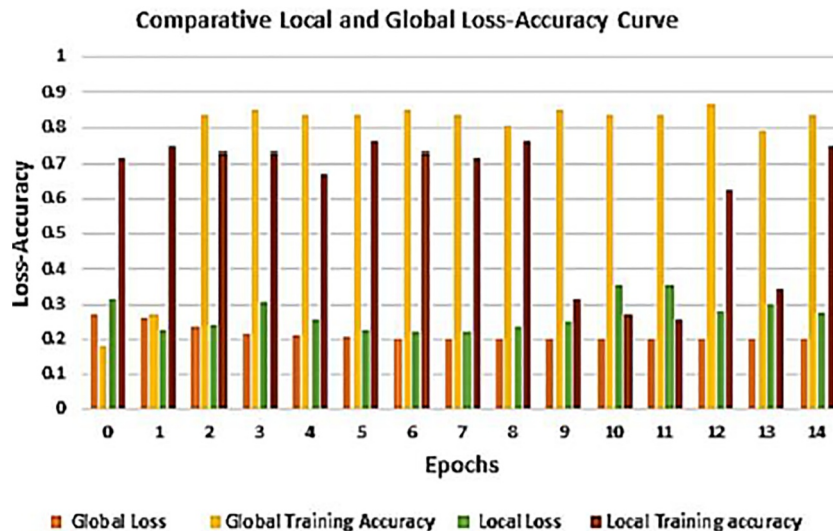


FIGURE 14 | Comparative local and global loss-accuracy curves.

aggregated global parameter set. This observation inspired leveraging over-the-air computation property [33] in the wireless multiple-access channel to perform aggregation directly without transmitting the individual local parameter sets to the server. This efficient and scalable federated learning approach can reduce the central server's computational burden and significantly reduce the communication bandwidth required, particularly in large-scale IoHT applications with numerous local clients. Several aspects of decentralised learning, such as communication efficiency, proper function estimation and data privacy, are crucial factors to take care of. In this scenario, the AirComp method is utilised for effective communication in decentralised learning. AirComp allows concurrent communication by multiple local nodes using the same wireless resources, leading to nonorthogonal communication and increased interference. Since the superposition property of the wireless channel is desired, this communication method can result in higher throughput than traditional orthogonal communication systems, with the gain scaling with the number of devices involved. This makes it a promising technique for practical systems, where high throughput is desired. In the following subsection, we provide a mathematical description of the AirComp model.

If n local nodes share n model gradients, then the information content y_i of the i th local node can be represented as in Equation (19),

$$y_i = x_i \quad (19)$$

where x_i is the i th model gradient.

In AirComp, simultaneous communication of multiple nodes over the same frequency band leads to the baseband representation of the received signal with multiple access channel (MAC). So, the signal received at the global node is modelled in Equation (20),

$$Y = \sum_{i=1}^n x_i \quad (20)$$

If the uplink fading coefficient of the wireless channel is h , Y becomes

$$Y = h \sum_{i=1}^n x_i \quad (21)$$

Considering z as the additive noise component of the channel due to interference of unknown signals,

$$Y = h \sum_{i=1}^n x_i + z \quad (22)$$

This is the weighted sum along with some additive noise.

Orthogonal communication ensures almost interference-free communication. As the reconstruction of individual messages is not required in a federated model creation, orthogonal communication is inefficient and wastes radio resources (time/frequency/space).

In an ideal scenario,

$$Y = \sum_{i=1}^n x_i \quad (23)$$

Thus, AirComp communication consumes only one block of radio resources. It requires n blocks of radio resources in the case of orthogonal communication. The improvement in efficiency level can be achieved by utilising two factors, smaller bandwidth and minimised latency.

AirComp allows for calculating a wide range of functions beyond just the weighted sum through the application of suitable pre- and post-processing functions. Thus, the desired pre-processing function is

$$f(x) = f_1\left(h \sum_{i=1}^n f_1(x_i) + \varepsilon\right) \quad (24)$$

Similarly, the desired post-processing function is

$$f(x) = f_2(Y) = f_2\left(h \sum_{i=1}^n x_i + \varepsilon\right) \quad (25)$$

Considering both pre- and post-processing,

$$f(x) = f_2(Y) = f_2\left(h \sum_{i=1}^n f_1(x_i) + \varepsilon\right) \quad (26)$$

4.8 | Privacy Preservation in Quantum Federated Learning (QFL) Architecture

Medically significant information about patients is confidential. Thus, preserving patient data privacy and security is essential. Several local data collection nodes, such as medical investigation and monitoring systems, provide data to train the local models in quantum computers. The job scheduling for quantum computers is based on the queueing delays and fidelity trends of the quantum machines. Each dictionary object containing trained model parameter information is received from the cluster of local nodes through the network to the global cloud base station (BS). The global model is broadcast to the local nodes. Test evaluation takes place on local patient data at the local nodes. This way, model efficiency is enhanced, but local data privacy is preserved. We enforce execution of this model following the proposed three-tier QFL network architecture design.

4.8.1 | Tier I: Sensor (Fundus Cameras)

Local hospitals gather patient data using Fundus cameras to form the local dataset. These collected data are transmitted over the access point to the local quantum cloud. Quantum models are trained on these local datasets.

4.8.2 | Tier II: Quantum Cloud (IBMQ/AWS Braket Quantum Hardware)

A number of quantum computers form the quantum compute clusters, where individual training takes place with local data. For successful model training, the PCA parameters or weights of the model are exchanged through epochs. In this way, several local models are created. These adjusted local model parameters are transmitted through the gateway to the global cloud for aggregation.

4.8.3 | Tier III: Global Aggregator Cloud

Responsible for the geospatial selection and aggregation of the local models. Local nodes adapt the aggregated model parameters to evaluate the aggregated quantum model. AirComp can estimate the uplink average of gradients.

The philosophy of AirComp enabled and layered ecosystem of the privacy-preserving quantum federated model learning framework for retinal image classification is presented in Figure 13.

5 | Discussion and Challenges

We implemented federated learning of the convolutional QNN/QQC model for retinal image patch classification and assessed its performance. The contributions of quantum learning are less evaluated for distributed privacy-preserving retinal image classification. The CNN model is coupled with quantum circuits to generate FedQCNN. Images are convoluted to obtain higher-level feature representations [50], which are reduced to smaller dimension vectors before converting classical information to quantum information and then passing quantum information through the quantum circuit. We implemented a 16-qubit circuit and achieved a promising test accuracy of 96.8%. Models may be trained either in quantum computers or simulators.

5.1 | Quantum Learning: Advantage Over Classical Learning

The advantage of implementing QNN is that the computationally exhaustive training phase consisting of several forward and backward passes may be executed with exponential speedup within the real quantum device. Also, according to some notable developments [2, 17, 32, 44], QNN has potentially better pattern generalisation capability with less amount of data. Light weight model with less number of learnable parameters provides the advantage of quantum model over the classical models.

5.2 | Distributed Learning: Advantage Over Local Learning

The geographical location-specific local models are learnt from medical data available in that region with consecutive forward

and backward training passes using the quantum gradient descent algorithm. Thus, local models are rich in knowledge extracted from diverse training data from various region-specific races, nations and countries containing multiple categories of disease symptoms, which helps to create an enriched global model. Furthermore, it is evident from experimental results that the proposed FedQCNN model can classify affected retinal patches effectively.

5.3 | Image Resizing: Compatibility With NSQ Datasets

We avoided direct image resizing to prevent medical information loss and performed experimentation with the image patches of size 16×16 . Smaller patch sizes are not able to encompass exudates adequately. These patches are passed through convolution layers to extract essential image features. The output vector of two convolutional and max pooling layers is of size 4×4 , which conforms to pass through the quantum opportunity in this NISQ era.

5.4 | Comparative Analysis: Classical Versus Quantum

The quantum circuit has 32 trainable model parameters. During epochs, these 32 weights are updated to minimise the training loss. We created a classical CNN model with the same convolution part. The summary of this CNN model with 205 parameters is shown in Figure 15. Here, each convolution layer has 4 filters. With the increasing number of filters, the classical model becomes efficient in capturing complex features, but the number of trainable parameters increases drastically. We received the comparable CNN test accuracy of 89.54% with 20 epochs.

5.5 | Comparative Analysis: Existing Literature

Table 2 shows the significant expressive power of QNN when compared with some state-of-the-art deep learning applications in the domain of affected retinal image identification [51, 52]. We provided a comparative performance of retinal classification work [15] with QNN in Table 3. In ref. [15], Mathur et al. implemented the QNN circuit in the classical neural network framework. They experimented with a reduced image size of 4

Layer (type)	Output Shape	Param #
conv2d_2 (Conv2D)	(None, 14, 14, 4)	40
max_pooling2d_2 (MaxPooling2D)	(None, 7, 7, 4)	0
conv2d_3 (Conv2D)	(None, 5, 5, 4)	148
max_pooling2d_3 (MaxPooling2D)	(None, 2, 2, 4)	0
flatten_1 (Flatten)	(None, 16)	0
dense_1 (Dense)	(None, 1)	17

Total params: 205 (820.00 B)

FIGURE 15 | Architecture of comparable classical CNN.

TABLE 3 | Comparison with deep learning and quantum learning-based medical image classification methods.

Method, year	Learning model	Medical image	Evaluation accuracy (%)
Deep learning applications			
Wang et al. (2020) [51]	Region-based fully convolutional network (R-FCN)	Retinal image	92.95
Alyoubi et al. (2021) [52]	CNN512 and YOLOv3	Retinal image	89
Proposed method	FedQCNN	Retinal image	96.8
Quantum learning applications			
Sengupta et al. (2021) [16]	Quantum neural network (QNN)	CT-scan	96.9
Hussein et al. (2022) [17]	Quantum neural network (QNN)	Chest X-Ray	98.1
Mathur et al. (2021) [15]	Quantum neural network (QNN)	Retinal fundus images	80
Proposed method	FedQCNN	Retinal image	96.8

Note: Boldface indicates the proposed method.

or 8 dimensions using both QNN and orthogonal QNN. Their results showed that the QNN accuracy is better than orthogonal QNN. Furthermore, they performed forward training and inference with their QNN. We compared the medical image classification works based on quantum learning together with the retinal image classification works based on classical deep learning in Table 3. However, the quantum classification works evaluate quantum learning-based classification tasks for identifying chest X-rays [17] and CT scan images [16] affected by COVID-19.

5.6 | Bandwidth and Latency Efficient Communication: AirComp

Although the distributed model optimisation algorithm has significant use in distributed ML [53], it is expensive in terms of communication resources because, for T iterations, n gradients are communicated per iteration, each consisting of k parameter updates, leading to a total of nkT scalars that must be communicated over the wireless channel. Though the data volume of model parameters is remarkably less than the raw data itself, nevertheless, wireless links are resource-hungry—a bottleneck in real-time implementation of federated learning models. In this configuration, wireless communication technologies such as the existing infrastructure of classical 5G cellular network links are used to send these local QCNN model parameters to the global node. This is a feasible solution to the decentralised quantum model learning without using quantum channels for transferring quantum data or objects to the global server. Implementation of a quantum communication network for distributed quantum model learning is a challenging point, and no work, as per our knowledge, previously implemented QFL with quantum channels. A prospective solution is the use of bandwidth and latency efficient over-the-air computation. Since the server does not use the individual model updates for any other purpose than aggregation, these model updates could be aggregated by over-the-air

computation (AirComp). AirComp promises to communicate with massive gains in energy, latency and spectrum efficiency compared to orthogonal communication methods. The expected efficiency of AirComp is due to the complete spectrum sharing for all participating devices. Unlike in traditional physical-layer communications, where interference is avoided by allocating orthogonal communication channels, AirComp promotes interference to compute a function of the individually transmitted messages.

AirComp ensures privacy in federated learning by allowing users to transmit only aggregated model updates rather than raw data. The updates are superimposed directly in the wireless channel, which prevents the server from accessing individual user contributions.

5.7 | Challenges of the Work and Potential Remedies

5.7.1 | Hybrid System Implementation

We implemented the federated learning of the QNN/PQC model for medical image classification and evaluated its performance. The quantum circuit can be augmented with a transfer learning model [54] to investigate the potential of knowledge transfer in the quantum learning domain. In the transfer learning association, the pre-trained knowledge in the form of reduced dimension vectors may be transferred to the circuit as quantum data.

5.7.2 | Advanced Circuit Design

Different parameterised circuits affect the model's landscapes. QNN can be designed and experimented with multi-qubit gates. Increasing qubits and higher circuit depth are heading towards the sophisticated circuit design. In that case, a proper gradient

estimator is also required for executing the QNN training cycle. This is an open door for further research advancement.

5.7.3 | PQC Gradient Estimation

For gradient-based learning, truncation and round-off error of finite difference method sometimes ill-estimate the gradients. In that case, circuit learning becomes unstable. The probable solution may be analytical gradient estimation. Jose and Simeone [55] provided an overview of gradient estimation and convergence analysis in the quantum computing paradigm. However, even gradient-based learning, like the noisy experimental environment, becomes unsuitable in some situations. In such conditions, gradient-free methods for circuit learning may be suitable. Particle swarm gradient-free optimisation [36, 56] and genetic algorithm-based gradient-free optimisation [57] may be adopted as potential solutions.

5.7.4 | Vanishing Gradient Problem

QNN uses a gradient descent optimisation algorithm; hence, it is required to avoid vanishing gradient or disappearance of the gradient problem during learning iterations. In the quantum learning domain, this vanishing gradient problem is called the ‘barren plateau’ problem [58, 59]. The probability of a ‘barren plateau’ increases exponentially with the number of qubits, and the phenomenon can be prevented in a small-scale circuit by initialising with suitable parameters. According to them, the mean of the circuit parameters is zero, and the variance decreases exponentially with increasing qubits. According to Grant et al. [59], a good choice of remedy to this problem may be to cut the whole circuit into several shallow blocks and initialise the network with some randomly selected parameters. Other parameters are chosen later. In this way, circuit depths can be controlled. However, researchers have yet to study the problem ‘barren plateau’.

5.7.5 | Adversarial Attack and Vulnerability of AirComp

The limitation of using federated learning architecture is the possibility of malicious attacks on the network. In the process, confidential information may be stolen in an unauthorised way. To address the challenge, we may use safety protocols to boost the network’s security. Differential privacy [9, 42] is a standard data protection policy to ensure data security and privacy. Again, integrating blockchain with federated learning to remove the central authority enables decentralised privacy protection [41].

AirComp may be susceptible to inference attacks, where the server infers individual updates in sparse user scenarios like knowledge about dominant user or small number of users etc. Malware-infected signal jamming and leakage of channel state information (CSI) can also compromise the system. Further, analogue computation lacks rollback or verification mechanisms, making the aggregation irreversible and sensitive to noise.

5.7.6 | Advanced Decentralised Learning

We proposed weighted average parameter aggregation while generating the global model. Different variations of aggregation methods or different federated learning algorithms may be experimented with to compare their performances. An alternative to the federated model is the gossip model. Gossip learning [60, 61] may be implemented without a central node, where learning occurs gradually by exchanging knowledge among the peer nodes. Further, one global aggregation server may be a bottleneck and a single point of failure. To overcome this limitation, we may introduce multiple aggregation clouds. These are the scopes for further research advancements.

6 | Conclusion

The paper addresses the ever-growing concerns of automated medical image diagnostics—medical data privacy maintenance and the requirement of huge annotated data for the successful training of time, memory and computation-hungry deep neural network. The facility to process this huge amount of data requires massive and condensed computation capability. Quantum machine learning is a new way of information processing, harnessing the principles of quantum computing to infer some decisions. In this work, multiple local quantum nodes train the model on local private data and collaborate to create a global quantum model. Over-the-air computation significantly reduces communication overhead, as multiple clients can simultaneously transmit their parameters to the central server using the same channel. Moreover, we can achieve significant benefits of distributed systems like-fault tolerance, reliability, scalability and efficiency through this federated learning model implementation. The solution can help to implement a high-speed computer-aided medical image diagnostic system. The system is capable of making decisions regarding the disease manifestation in retinal images and, at the same time, preserving the privacy of patient data. If the PQC model is properly scaled with the real-life application, it may emerge as an efficient automated application for near-term devices [49]. It will extend a new horizon to the evolution and application of quantum learning with wireless networks in a popular healthcare research domain.

Author Contributions

Mahua Nandy Pal: conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, resources, software, validation, visualization, writing – original draft, writing – review and editing. **Debashis De:** data curation, formal analysis, funding acquisition, investigation, project administration, resources, supervision, validation, visualization.

Acknowledgements

The authors acknowledge the contributions of the Meity QCAL project, Ministry of Electronics and Information Technology, Government of India, (Meity/AWS/Cohort 2/2023 dated 25th April, 2023) for funding this research and AWS for the Amazon Braket execution interface.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data are available from the online public repository: E-optha Dataset, A consortium of Institutes and Hospitals, France, Available: <http://www.adcis.net/en/third-party/e-optha/>.

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